📊 Exploratory Data Analysis (EDA) Summary

📌 1 Overview

This dataset contains sales transaction records from different regions, sales representatives, and product categories. The purpose of this analysis is to understand sales patterns, customer behavior, product performance, and regional trends — ultimately to support business decisions and sales strategy improvements.

📌 2 Key Findings

Top Product Categories:

Furniture and Electronics have the highest total sales amounts.

High-Performing Regions:

The North and West regions generate higher average sales per transaction.

Best Sales Reps:

Bob is the top-performing sales rep, covering multiple regions effectively.

Customer Type Trends:

Returning customers tend to purchase higher quantities and spend more on average compared to new customers.

Sales Channel Patterns:

Retail sales generally contribute to higher quantities sold, while Online sales tend to have higher average sale values.

📌 3 Visual Support

Bar Chart: Product Category vs. Total Sales

Pie Chart: Sales split by Region

Heatmap: Correlation between Sales_Amount, Unit_Price, Quantity_Sold, and Discount

Boxplot: Sales_Amount by Customer Type

Countplot: Number of transactions by Payment Method

📌 4 Business Implications

Focus marketing and promotions on Furniture and Electronics, especially in North and West regions.

Retaining returning customers should be a priority, as they bring in more revenue.

Bob's performance strategies could be analyzed and replicated for other sales reps.

Increase emphasis on Online sales channels for higher-value transactions.

```python
## Import libraries
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```python
## Load the dataset
data = pd.read_csv('C:/Users/Abhijeet Kuanr/Downloads/Elevate Labs/Task-5/sales_data.csv')
data.head()
```

	Product_ID	Sale_Date	Sales_Rep	Region	Sales_Amount	Quantity_Sold	Product_Category	Unit_Cost	Unit_Price	Customer_Type	Discount	Payment_Method	Sales_Channel	Region_and_Sales_Rep
0	1052	2023-02-03	Bob	North	5053.97	18	Furniture	152.75	267.22	Returning	0.09	Cash	Online	North-Bob
1	1093	2023-04-21	Bob	West	4384.02	17	Furniture	3816.39	4209.44	Returning	0.11	Cash	Retail	West-Bob
2	1015	2023-09-21	David	South	4631.23	30	Food	261.56	371.40	Returning	0.20	Bank Transfer	Retail	South-David
3	1072	2023-08-24	Bob	South	2167.94	39	Clothing	4330.03	4467.75	New	0.02	Credit Card	Retail	South-Bob
4	1061	2023-03-24	Charlie	East	3750.20	13	Electronics	637.37	692.71	New	0.08	Credit Card	Online	East-Charlie

```python
# Basic information about the dataset
print(data.info())
print(data.shape)

# Summary statistics of the dataset
print(data.describe())
```

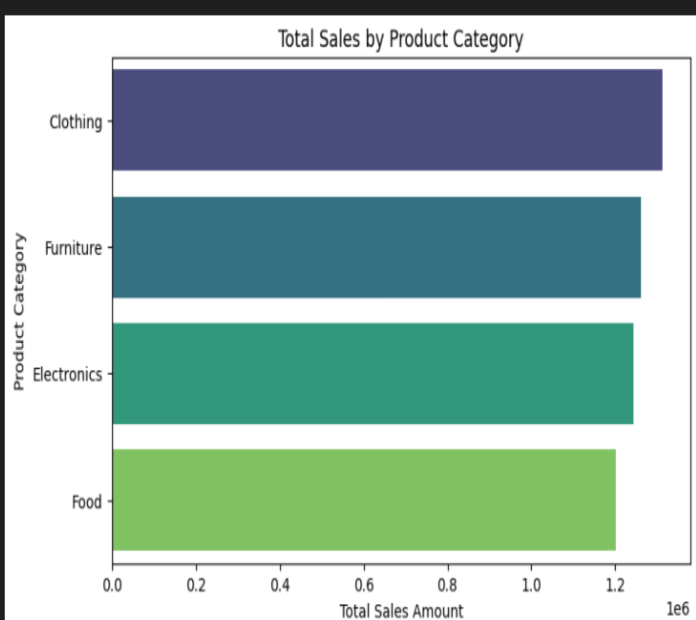
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 14 columns):
 #   Column                Non-Null Count  Dtype
---  ------                --------------  -----
 0   Product_ID            1000 non-null   int64
 1   Sale_Date             1000 non-null   object
 2   Sales_Rep             1000 non-null   object
 3   Region                1000 non-null   object
 4   Sales_Amount          1000 non-null   float64
 5   Quantity_Sold         1000 non-null   int64
 6   Product_Category      1000 non-null   object
 7   Unit_Cost             1000 non-null   float64
 8   Unit_Price            1000 non-null   float64
 9   Customer_Type         1000 non-null   object
 10  Discount              1000 non-null   float64
 11  Payment_Method        1000 non-null   object
 12  Sales_Channel         1000 non-null   object
 13  Region_and_Sales_Rep  1000 non-null   object
dtypes: float64(4), int64(2), object(8)
memory usage: 109.5+ KB
None
(1000, 14)
        Product_ID  Sales_Amount  Quantity_Sold    Unit_Cost   Unit_Price  \
count  1000.000000   1000.000000    1000.000000  1000.000000  1000.000000
...
25%        0.08000
50%        0.15000
75%        0.23000
max        0.30000
```

```python
# Check for missing values
print(data.isnull().sum())
```

```
Product_ID             0
Sale_Date              0
Sales_Rep              0
Region                 0
Sales_Amount           0
Quantity_Sold          0
Product_Category       0
Unit_Cost              0
Unit_Price             0
Customer_Type          0
Discount               0
Payment_Method         0
Sales_Channel          0
Region_and_Sales_Rep   0
dtype: int64
```

```python
# 1 Product Category vs. Total Sales
plt.figure(figsize=(8,5))
category_sales = data.groupby('Product_Category')['Sales_Amount'].sum().sort_values(ascending=False)
sns.barplot(x=category_sales.values, y=category_sales.index, palette='viridis')
plt.title('Total Sales by Product Category')
plt.xlabel('Total Sales Amount')
plt.ylabel('Product Category')
plt.show()
```

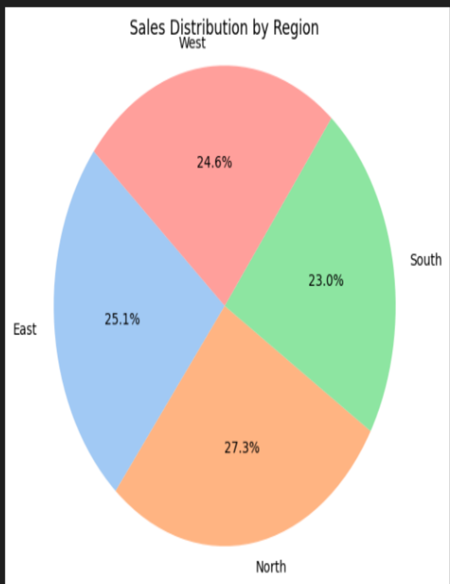
```
C:\Users\Abhijeet Kuanr\AppData\Local\Temp\ipykernel_25680\2059589606.py:4: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `y` variable to `hue` and set `legend=False` for the same effect.

  sns.barplot(x=category_sales.values, y=category_sales.index, palette='viridis')
```



1. **Total Sales by Product Category**: A bar chart visualizing the total sales amount for each product category. It highlights which categories (e.g., Clothing, Furniture, Electronics, Food) contribute the most to overall sales.

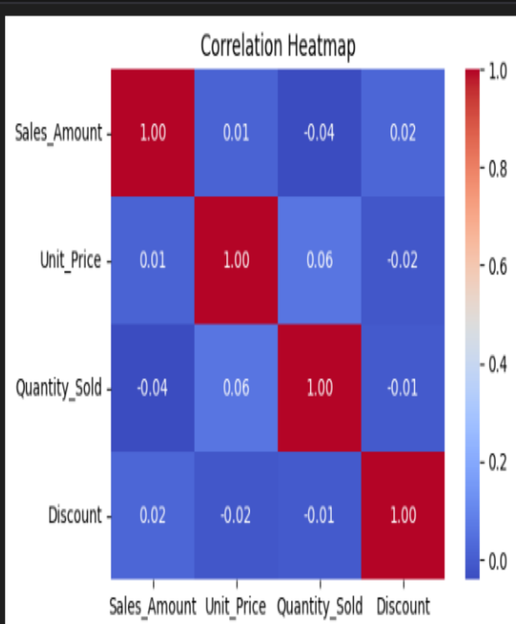
```python
# 2 Sales Split by Region (Pie Chart)
region_sales = data.groupby('Region')['Sales_Amount'].sum()
plt.figure(figsize=(6,6))
plt.pie(region_sales, labels=region_sales.index, autopct='%1.1f%%', startangle=140, colors=sns.color_palette('pastel'))
plt.title('Sales Distribution by Region')
plt.axis('equal')
plt.show()
```

Python



2. **Sales Distribution by Region**: A pie chart showing the percentage split of sales across different regions (East, North, South, West). This helps identify regional performance and market share. From the chart you can clearly see North region have higher Distribution.

```python
# 3 Correlation Heatmap
plt.figure(figsize=(6,4))
numeric_cols = ['Sales_Amount', 'Unit_Price', 'Quantity_Sold', 'Discount']
sns.heatmap(data[numeric_cols].corr(), annot=True, cmap='coolwarm', fmt=".2f")
plt.title('Correlation Heatmap')
plt.show()
```



3. **Correlation Heatmap**: A heatmap displaying the correlation between numeric variables such as `Sales_Amount`, `Unit_Price`, `Quantity_Sold`, and `Discount`. It provides insights into relationships between these variables.

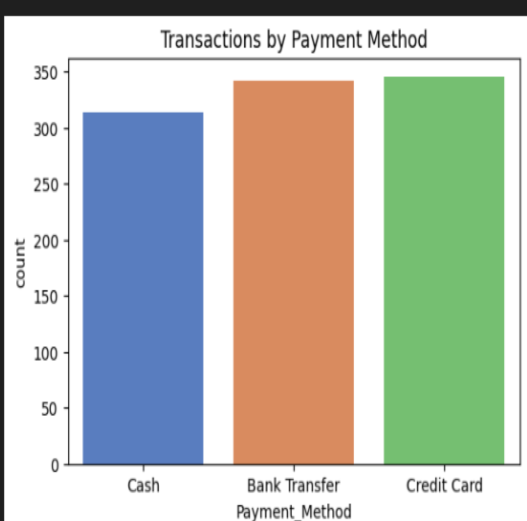
```python
# 4 Boxplot – Sales by Customer Type
plt.figure(figsize=(6,4))
sns.boxplot(data=data, x='Customer_Type', y='Sales_Amount', palette='Set2')
plt.title('Sales Amount by Customer Type')
plt.show()
```

```
C:\Users\Abhijeet Kuanr\AppData\Local\Temp\ipykernel_25680\592959886.py:3: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

  sns.boxplot(data=data, x='Customer_Type', y='Sales_Amount', palette='Set2')
```

4. **Sales Amount by Customer Type**: A boxplot comparing the distribution of sales amounts for different customer types (e.g., New vs. Returning). It highlights variations in spending behavior.

```python
# 5 Countplot – Number of Transactions by Payment Method
plt.figure(figsize=(6,4))
sns.countplot(data=data, x='Payment_Method', palette='muted')
plt.title('Transactions by Payment Method')
plt.show()
```

```
C:\Users\Abhijeet Kuanr\AppData\Local\Temp\ipykernel_25680\1813617839.py:3: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

  sns.countplot(data=data, x='Payment_Method', palette='muted')
```



5. **Transactions by Payment Method**: A countplot showing the number of transactions for each payment method (e.g., Cash, Credit Card, Bank Transfer). It helps understand customer preferences for payment.